

Exact 1st order ODE

$$F(x, y) = C \Leftrightarrow dF = \left[\frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy = 0 \right]$$

solution of $\Rightarrow$

Exact Differential Form

Definition 2. The differential form $M(x, y)dx + N(x, y)dy$ is said to be **exact** in a rectangle R if there is a function $F(x, y)$ such that

$$(4) \quad \frac{\partial F}{\partial x}(x, y) = M(x, y) \quad \text{and} \quad \frac{\partial F}{\partial y}(x, y) = N(x, y)$$

for all (x, y) in R . That is, the total differential of $F(x, y)$ satisfies

$$dF(x, y) = M(x, y)dx + N(x, y)dy .$$

If $M(x, y)dx + N(x, y)dy$ is an exact differential form, then the equation

$$M(x, y)dx + N(x, y)dy = 0 \quad \Rightarrow \quad F(x, y) = C$$

is called an **exact equation**.

How to determine that ODE is exact?

Test for Exactness

Theorem 2. Suppose the first partial derivatives of $M(x, y)$ and $N(x, y)$ are continuous in a rectangle R . Then

$$M(x, y)dx + N(x, y)dy = 0$$

is an exact equation in R if and only if the compatibility condition

$$(5) \quad \frac{\partial M}{\partial y}(x, y) = \frac{\partial N}{\partial x}(x, y)$$

holds for all (x, y) in R .[†]

$$M = \frac{\partial F}{\partial x}, \quad N = \frac{\partial F}{\partial y}$$

$$\frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x}$$

$$\boxed{\frac{\partial}{\partial x} N = \frac{\partial}{\partial y} M}$$

$$\underbrace{(x+y)}_{\text{"M"}} dx + \underbrace{(x-y)}_{\text{"N"}} dy = 0$$

$$\frac{\partial}{\partial y}(x+y) = 1, \quad \frac{\partial}{\partial x}(x-y) = 1 \Rightarrow \text{exact}$$

Method for Solving Exact Equations

- (a) If $M dx + N dy = 0$ is exact, then $\partial F / \partial x = M$. Integrate this last equation with respect to x to get

$$(11) \quad F(x, y) = \int M(x, y) dx + g(y) .$$

- (b) To determine $g(y)$, take the partial derivative with respect to y of both sides of equation (11) and substitute N for $\partial F / \partial y$. We can now solve for $g'(y)$.
 (c) Integrate $g'(y)$ to obtain $g(y)$ up to a numerical constant. Substituting $g(y)$ into equation (11) gives $F(x, y)$.
 (d) The solution to $M dx + N dy = 0$ is given implicitly by

$$F(x, y) = C$$

(Alternatively, starting with $\partial F / \partial y = N$, the implicit solution can be found by first integrating with respect to y ; see Example 3.)

$(x+y)dx + (x-y)dy = 0$
 to solve, we need
 to find $F(x, y)$ s.t.

$$\begin{cases} \frac{\partial F}{\partial x} = x+y \\ \frac{\partial F}{\partial y} = x-y \end{cases} \Rightarrow$$

$$\Rightarrow F(x, y) = \frac{x^2}{2} + yx + f(y)$$

$$\Rightarrow \frac{\partial F}{\partial y} = [0 + x + f'(y)] = x - y \Rightarrow f'(y) = -y \Rightarrow$$

$$\Rightarrow f(y) = -\frac{y^2}{2} + \cancel{C_1} \Rightarrow F(x, y) = \frac{x^2}{2} + yx - \frac{y^2}{2} + \cancel{C_1}$$

Solution: $\frac{x^2}{2} + xy - \frac{y^2}{2} + \cancel{C_1} = C \leftarrow \text{implicit solution}$

Example 1 Solve the differential equation

$$\frac{dy}{dx} = -\frac{2xy^2 + 1}{2x^2y}.$$

not separable
not linear

$$2x^2y dy = -(2xy^2 + 1) dx \Rightarrow \underbrace{(2xy^2 + 1)}_{\frac{\partial}{\partial y}(2xy^2 + 1)} dx + \underbrace{2x^2y}_{\frac{\partial}{\partial x}(2x^2y)} dy = 0$$

Is it exact?

$$\frac{\partial}{\partial y}(2xy^2 + 1) = 4xy, \quad \frac{\partial}{\partial x}(2x^2y) = 4xy$$

$\Rightarrow$ yes, exact.

Solve:

$$\begin{cases} \frac{\partial F}{\partial x} = 2xy^2 + 1 \\ \frac{\partial F}{\partial y} = 2x^2y \end{cases} \Rightarrow F = x^2y^2 + f(x), \quad \frac{\partial F}{\partial x} = \boxed{2xy^2 + f'(x) = 2xy^2 + 1}$$

$$\Rightarrow f'(x) = 1 \Rightarrow f(x) = x \Rightarrow F(x, y) = x^2y^2 + x$$

$$\text{Solution: } x^2y^2 + x = C$$

Example 3 Solve

not separable
linear
(simpler)

(15)

$$(1 + e^x y + x e^x y) dx + (x e^x + 2) dy = 0.$$

$$1 + (e^x + x e^x) y + (x e^x + 2) \frac{dy}{dx} = 0$$

Is it exact? $\frac{\partial}{\partial y} \underline{\quad} = \underline{e^x + x e^x}$, $\frac{\partial}{\partial x} \underline{\quad} = \underline{e^x + x \cdot e^x}$

$$\begin{cases} \frac{\partial F}{\partial x} = 1 + e^x y + x e^x y \\ \frac{\partial F}{\partial y} = x e^x + 2 \end{cases} \Rightarrow F = x e^x y + 2y + f(x) \Rightarrow \text{exact}$$

$$\frac{\partial F}{\partial x} = e^x y + x e^x y + f'(x) = 1 + e^x y + x e^x y \Rightarrow f'(x) = 1$$

$$\Rightarrow F = x e^x y + 2y + x$$

$$\Rightarrow \boxed{x e^x y + 2y + x = C}$$

$$y = \frac{C - x}{x e^x + 2}$$

Example 4 Show that

(16) $(x + 3x^3 \sin y)dx + (x^4 \cos y)dy = 0$

is *not* exact but that multiplying this equation by the factor x^{-1} yields an exact equation. Use this fact to solve (16).

Is it exact? $\frac{\partial}{\partial y} \underline{\quad} = \cos y \cdot 3x^3$ $\frac{\partial}{\partial x} \underline{\quad} = 4x^3 \cos y$
not exact

$\cdot x^{-1} \Rightarrow (1 + 3x^2 \sin y)dx + (x^3 \cos y)dy = 0$
 $\frac{\partial}{\partial y} \downarrow = 3x^2 \cos y$ $\frac{\partial}{\partial x} \downarrow = 3x^2 \cos y \Rightarrow \text{exact}$

$\begin{cases} \frac{\partial F}{\partial x} = 1 + 3x^2 \sin y \\ \frac{\partial F}{\partial y} = x^3 \cos y \end{cases} \Rightarrow F = x + x^3 \sin y + f(y)$
 $\frac{\partial F}{\partial y} = x^3 \cos y + f'(y) = x^3 \cos y$
so $f'(y) = 0 \Rightarrow f(y) = C = 0$
 $\Rightarrow F(x, y) = x + x^3 \sin y$
 $\Rightarrow \boxed{x + x^3 \sin y = C}$